



Integral Energy's Annual Network Prices Report

April 2008

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1 Introduction

Under Clause 12(5) of the Independent Pricing And Regulatory Tribunal's (IPART) Network Price Determination released in June 2004 (2004 Network Determination), Integral is required for each year of the Regulatory Control period to submit to the Tribunal the following information:

- Annual Network Pricing Proposal containing all of the information set out in Annexure 16 of the Draft Network Price Determination; and
- Annual Network Pricing Report containing all of the information set out in Annexure 17 of the Draft Network Price Determination.

Under Annexure 17 of the 2004 Network Determination released in June 2004, Integral is required to submit to the Tribunal an annual network pricing report that contains the following information:

- List of proposed prices for network tariffs and miscellaneous charges and monopoly fees;
- Discuss the forthcoming changes in network tariffs, the prices, structure or associated criteria, and any new or obsolete network tariffs, and the reasons for the change;
- Explain how the prices meet the regulatory arrangements, including price limits on network tariffs, weighted average price cap formula, and provide for the recovery of Transmission Related Payments;
- Demonstrate the impact of the forthcoming change in the network tariffs on typical customer's bills, including disclosing forecast average prices(based on typical bill categories);
- Confirm and demonstrate that the prices are consistent with the Pricing Principles and the information contained in the Network Tariff Strategy Statement;
- Outline the expected levels of service and projected capital expenditure projects that will occur for the coming year; and
- Outline the expected consumption for major customer classes and maximum demand for the coming year.

This report is intended to satisfy the requirements of Annexure 17 of the 2004 Network Determination.

2 Background

2.1 *Integral's network*

Integral's network serves over 2.1 million people in the Blue Mountains, Western Sydney, the Southern Highlands and the Illawarra. In terms of customer connections, this involves the provision of electricity to approximately 750,000 residential customers and 75,000 commercial/industrial customers.

Integral distributes electricity taken from the NSW transmission network at either 132,000 or 66,000 volts. It is then reduced to 33,000, 22,000 or 11,000 volts throughout a network of subtransmission and zone substations. Street-based distribution substations reduce voltage to levels that meet Integral's customers' home or business requirements.

Key load centres within Integral's supply area include: Parramatta, Wollongong, Campbelltown, Blacktown and Liverpool.

2.2 *Integral's customers' characteristics*

Some of Australia's fastest growing communities (in terms of population and load) are located within Integral's network area. Major drivers of this growth include new residential release areas and the upsurge in air conditioning penetration. Due to rapid development, large parts of Integral's network are being transformed from the semi-rural character that underpinned its initial design and development, to a much more urban-based network. This is evident in new release areas in the north-west, including Kellyville and Parklea. This development, combined with ongoing urban consolidation, is creating pressures to raise network reliability performance from levels accepted as appropriate for rural areas, to levels expected by customers in urban areas.

The uptake of air conditioning within Western Sydney is also placing immense strain on Integral's network. Ten years ago, about one in four of households within Integral's supply area had air conditioning. Today, around three quarters of all households in Western Sydney have air conditioning and indications are that this figure will continue to grow.

Air conditioning places a huge strain on Integral's electricity network during summer months. A characteristic of air conditioning in the Western Sydney climate is that the appliances place significant demands on the network, but are usually required on a relatively small number of days. Air conditioners therefore have contributed to a significant growth in peak demand while total energy consumption growth has been slightly more modest.

The following chart highlights the dramatic impact of volatile air conditioning load on the daily consumption profile of a representative sample of residential customers in the Western Sydney Region.

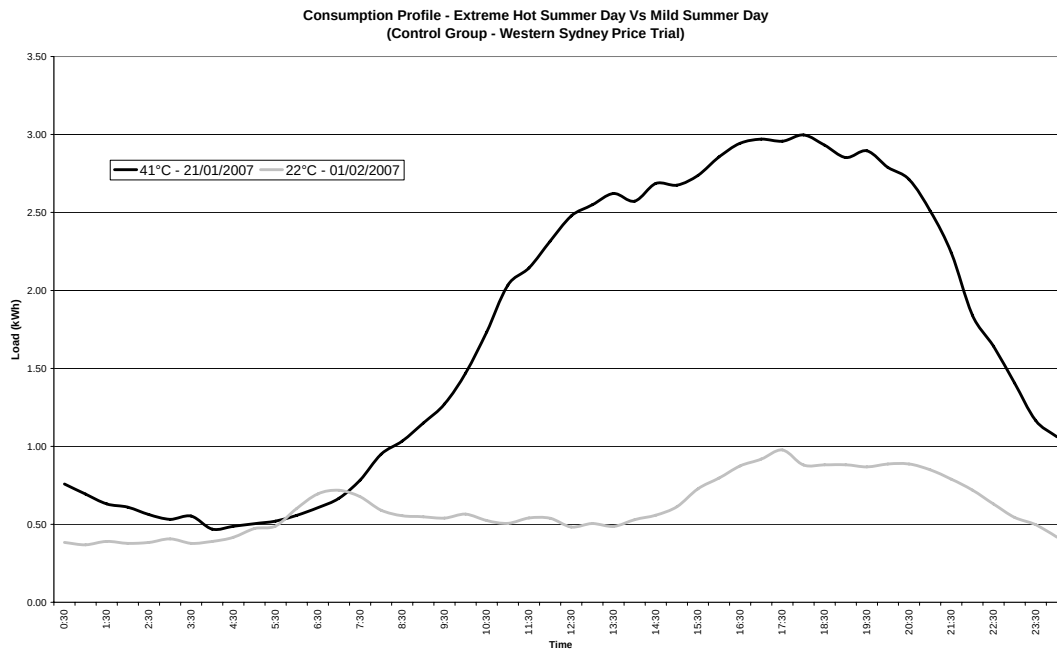


Figure 2-1: Impact of Volatile Residential Air Conditioning Load

To ensure that service and reliability standards are met and that customer service is not compromised, Integral will need to continue to invest heavily over the current regulatory period on growth and refurbishment related capital. The historical development of Integral's network has resulted in significant numbers of major system assets currently reaching the end of their effective service lives, requiring increased capital expenditure to replace these assets and ensure customer service levels are maintained.

3 Proposed Network Tariff Reforms

3.1 Background

Integral is committed to undertaking structural network tariff reform and introducing new network tariffs where there are economic benefits to be realised. Integral also understands that network tariff reform can, if not carefully implemented, result in customers being exposed to material adjustment costs. Integral believes that it is important to provide customers and retailers with advance notice of its intentions in relation to network tariff strategy. This will ensure that customers affected by these reforms are well placed to mitigate their impact by responding to the tariff incentives in the appropriate manner.

Integral has undertaken substantial network tariff reform over the past five years, as summarised below:

- Introduction of the Inclining Block Tariff (IBT) for residential and general supply customers in 2004/05 with the price difference between the two consumption blocks having been increased from 5% in 2004/05 to 20% in 2007/08;
- Integral has also reduced the annual consumption threshold under the IBT structure for general supply customers from 21,000 kWh in 2004/05 to 10,000 kWh in 2007/08;
- Introduction of new voluntary time of use tariffs to provide customers with annual consumption below 160 MWh pa with more tariff choices;
- A change to the time of use (Peak, Shoulder and Off-peak) period definitions for energy consumption to better reflect the summer peaking characteristics of Integral's network;
- Compulsory assignment of general supply customers with annual consumption in excess of 160 MWh pa to Integral's Low Voltage TOU Demand Tariff;
- Changed the structure of Integral's maximum kVA demand tariff to a peak-period basis in 2005/06 and then to a seasonal peak-period basis in 2007/08;
- Introduction of a new transitional network tariff in 2007/08 to facilitate the transition of general supply customers with annual consumption above 160 MWh pa to the low voltage TOU demand tariff;
- Increase to the peak energy price for all tariffs with a time of use energy component and reduce the shoulder energy price in line with the aim of eliminating the shoulder period over the longer term; and
- increase to the fixed charge of all network tariffs by more than the inflation rate, where it is economically appropriate to do so.

3.2 Proposed Network Tariff Reforms in 2008/09

The only major network tariff reform proposed to be undertaken in 2008/09 is to compulsorily assign approximately 470 customers with annual consumption in excess of 160 MWh pa from the general supply tariff to the transitional network tariff on 1 July 2008.

This reform proposal is consistent with Integral's pricing policy of compulsory demand pricing for all customers with annual consumption greater than 160 MWh, which was introduced after significant public consultation on 1 July 2004.

3.2.1 Proposed Transitional Arrangements

Integral is aware that moving customers from a basic tariff to a demand tariff has the potential to result in material changes (positive or negative) to the network bill of individual customers. To ensure that individual customers are not exposed to unacceptable network price shocks, Integral developed a comprehensive transitional strategy, as summarised below:

- To immediately assign an individual customer with annual consumption above 160 MWh pa to a the low voltage TOU demand tariff where it is reasonable to expect that the customer will receive a reduction or a moderate increase in their network bill from being assigned to a demand tariff.
- To assign an individual customer with annual consumption above 160 MWh pa to a transitional network tariff where it is reasonable to expect that the customer will receive an unacceptable increase in their network bill if they were assigned immediately to the low voltage TOU demand tariff. It is Integral's intention to compulsorily assign customers on the transitional tariff to the low voltage TOU demand tariff over a reasonable period

3.2.2 Estimated Customer Impact of Proposed Compulsory Assignment

Based on an analysis of historical consumption data, Integral expects that the customers assigned to the transitional network tariff will be exposed to the following annual network bill impact outcomes:

- Around two thirds of the customers assigned to the transitional network tariff are estimated to receive a network bill in 2008/09 that is lower than what it would have been if these customers had remained on their original network tariff.
- Around 25% of the customers assigned to the transitional network are estimated to receive a network bill in 2008/09 that is no more than 10% higher than what they are estimated to pay if they had remained on their original network tariff.
- The remaining 9% of the customers assigned to the transitional network are estimated to receive a network bill in 2008/09 that is more than 10% higher than what they are estimated to pay if they had remained on their original network tariff.
-

The following chart provides an indication of the annual network bill impact distribution of assigning approximately 470 customers to the transitional general supply time of use network tariff on 1 July 2008.

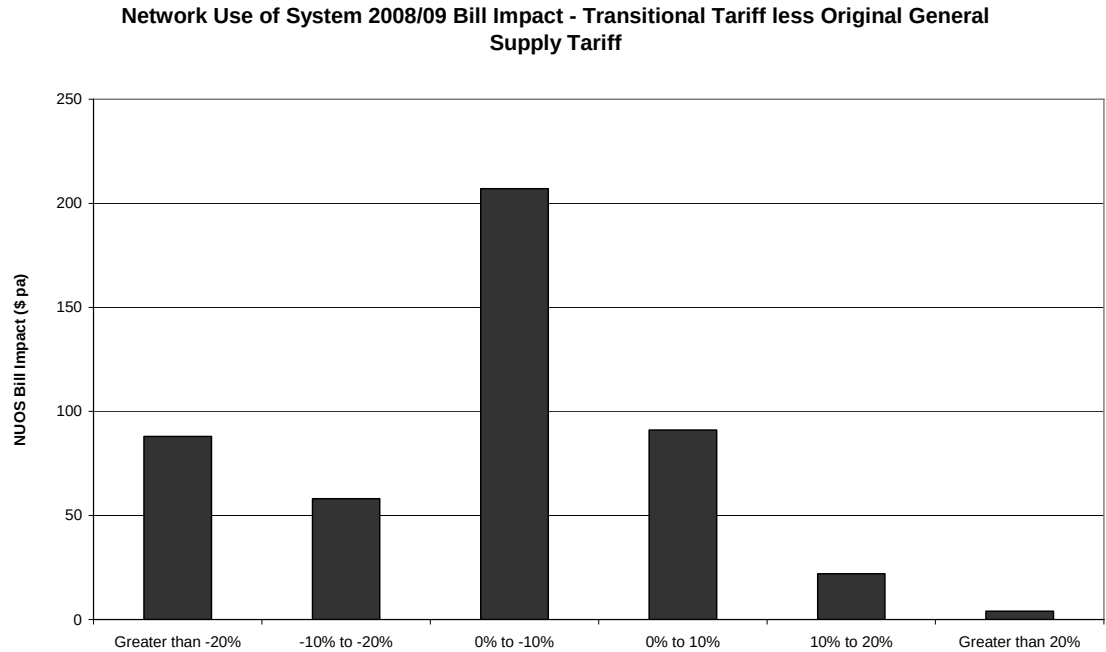


Figure 3-1: Annual NUOS Bill Impact Distribution – Transitional Tariff

Integral believes that it is possible that the actual network bill impact will be lower than represented in the above figure as some customers will respond to these new price signals by reducing their consumption during the peak period.

3.2.3 *Network Revenue Impact of Proposed Transitional Network Tariff*

Integral believes that the proposed compulsory assignment of these customers to the transitional network tariff will not have a material impact on Integral’s network revenue in 2008/09 given the relatively small number of customers involved. Nevertheless, it is possible that customers assigned to the transitional tariff will respond by reducing their energy consumption, particularly during peak periods, which will have a negative impact on Integral distribution revenue in 2008/09.

4 Proposed Network Prices for 2008/09

Annexure 17 of the 2004 Network Determination requires that the Annual Network Pricing Report submitted by a DNSP under clause 12 must:

List proposed prices for network tariffs and miscellaneous service charges and monopoly service charges.

This Section provides Integral's proposed 2008/09 prices for network tariffs and miscellaneous service charges and monopoly service charges¹.

4.1 Residential Tariffs

4.1.1 Default Residential Tariff

The following table shows the proposed prices for the Default Residential Tariff for 2008/09.

Component	2007/08		2008/09		% Change in DUOS	% Change in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$/pa)	76.65	76.65	76.65	76.65	0.0%	0.0%
First Block Charge (c/kWh)	5.37	5.96	5.73	6.56	6.7%	10.0%
Second Block Charge (c/kWh)	6.45	7.04	6.86	7.68	6.3%	9.1%
Block Threshold (kWh pa)	7,000	7,000	7,000	7,000		

Table 4-1: Integral's proposed 2008/09 Residential Inclining Block Tariff

¹ All rates quoted in this Section have been rounded to 2 decimal places and excludes Integral Energy's contribution to the NSW Governments Climate Change Fund.

4.1.2 Controlled Load Tariffs

The following table shows the proposed prices for Controlled Load 1 Network Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	7.30	7.30	7.30	7.30	0.0%	0.0%
Uniform Variable Charge (c/kWh)	0.19	0.47	0.19	0.52	0.0%	10.2%

Table 4-2: Integral's Proposed 2008/09 Controlled Load 1 Tariff

The following table shows the proposed prices for Controlled Load 2 Network Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	7.30	7.30	7.30	7.30	0.0%	0.0%
Uniform Variable Charge (c/kWh)	1.49	1.88	1.49	2.02	0.0%	7.6%

Table 4-3: Integral's Proposed 2008/09 Controlled Load 2 Tariff

4.1.3 Residential Time of Use (Type 5) Tariff

The following table shows the proposed prices for Residential Time of Use (Type 5) Network Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	200.75	200.75	208.05	208.05	3.6%	3.6%
Peak Consumption Charge (c/kWh)	8.76	9.58	7.04	7.78	-19.6%	-18.8%
Shoulder Consumption Charge (c/kWh)	6.04	6.86	7.04	7.78	16.7%	13.4%
Off-Peak Consumption Charge (c/kWh)	2.00	2.28	3.99	4.72	100.1%	107.2%

Table 4-4: Integral's Proposed 2008/09 Residential ToU (Type 5) Tariff

4.1.4 Residential Time of Use Tariff

The proposed prices for Integral's residential Time of Use (Type 6) Network Tariff for 2008/09 are shown in the following table.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	83.95	83.95	113.95	113.95	35.7%	35.7%
Peak Consumption Charge (c/kWh)	9.36	10.18	10.34	11.07	10.5%	8.7%
Shoulder Consumption Charge (c/kWh)	5.93	6.75	6.50	7.24	9.6%	7.2%
Off-Peak Consumption Charge (c/kWh)	1.62	1.90	1.81	2.54	11.9%	33.8%

Table 4-5: Integral's Proposed 2008/09 Residential Time of Use Tariff

4.2 Non-Residential Tariffs

4.2.1 Default General Supply Tariff

The following table provides a comparison of the actual network prices for 2007/08 and the proposed prices for the Default General Supply Network Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	76.65	76.65	76.65	76.65	0.0%	0.0%
First Block Charge (c/kWh)	4.33	4.88	4.66	5.34	7.8%	9.5%
Second Block Charge (c/kWh)	5.10	5.65	5.47	6.15	7.3%	8.9%
Block Threshold (kWh pa)	10,000	10,000	10,000	10,000		

Table 4-6: Integral's Proposed 2008/09 General Supply Inclining Block Tariff

4.2.2 General Supply Time-of-Use (Type 5) Tariff

The following table provides a comparison of the actual network prices for 2007/08 and the proposed prices for the General Supply Time of Use (Type 5) Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	200.75	200.75	208.05	208.05	3.6%	3.6%
Peak Consumption Charge (c/kWh)	8.76	9.58	5.57	6.54	-36.3%	-31.7%
Shoulder Consumption Charge (c/kWh)	6.84	7.66	5.57	6.54	-18.5%	-14.6%
Off-Peak Consumption Charge (c/kWh)	2.00	2.28	3.93	4.89	96.8%	114.6%

Table 4-7: Integral's Proposed 2008/09 General Supply ToU (Type 5) Tariff

4.2.3 General Supply Time-of-Use Tariff

The following table provides a comparison of the actual network prices for 2007/08 and the proposed prices for the General Supply Time of Use Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	124.10	124.10	127.75	127.75	2.9%	2.9%
Peak Charge (c/kWh)	7.62	8.44	8.40	9.10	10.1%	7.8%
Shoulder Charge (c/kWh)	5.34	6.16	5.84	6.54	9.4%	6.1%
Off-Peak Charge (c/kWh)	1.21	1.49	1.35	2.05	11.9%	37.8%

Table 4-8: Integral's Proposed 2008/09 General Supply Time of Use Tariff

4.2.4 Transitional Time-of-Use Tariff

The following table provides a comparison of the actual network prices for 2007/08 and the proposed prices for the Transitional Time of Use Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	2,482	2,482	2,613.4	2,163.4	5.3%	5.3%
Peak Charge (c/kWh)	7.68	8.50	8.46	9.24	10.2%	8.7%
Shoulder Charge (c/kWh)	4.09	4.91	4.44	5.22	8.6%	6.3%
Off-Peak Charge (c/kWh)	0.02	0.30	0.02	0.63	10.9%	109.2%

Table 4-9: Integral's Proposed 2008/09 Transitional Time of Use Tariff

4.2.5 Published Network Demand Tariffs

The following table provides a comparison of the actual network prices for 2007/08 and the proposed prices for the LV Time of Use Demand Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	2,482	2,482	2,613	2,613	5.3%	5.3%
Peak Charge (c/kWh)	1.47	2.28	1.50	2.15	2.6%	-5.7%
Shoulder Charge (c/kWh)	0.72	1.53	0.66	1.31	-7.3%	-14.6%
Off-Peak Charge (c/kWh)	0.01	0.29	0.01	0.51	5.8%	76.7%
Seasonal Peak Demand (\$/KVA/Month)	7.43	7.43	8.32	8.60	11.9%	15.7%
Non Seasonal Peak Demand (\$/KVA/Month)	7.08	7.08	7.92	8.20	11.9%	15.8%

Table 4-10: Integral's Proposed 2008/09 LV Time of Use Demand Tariff

Table 4-11 provides a comparison of the actual network prices for 2007/08 and the proposed prices for the HV ToU Demand Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	4,106	4,106	4,325	4,325	5.3%	5.3%
Peak Charge (c/kWh)	0.88	1.67	0.93	1.53	5.2%	-8.4%
Shoulder Charge (c/kWh)	0.44	1.22	0.43	1.02	-1.7%	-16.5%
Off-Peak Charge (c/kWh)	0.01	0.28	0.01	0.47	5.8%	66.9%
Seasonal Peak Demand (\$/KVA/Month)	5.31	5.31	5.94	6.21	11.9%	16.9%
Non Seasonal Peak Demand (\$/KVA/Month)	5.06	5.06	5.66	5.92	11.9%	17.0%

Table 4-11: Integral's Proposed 2008/09 HV Time of Use Demand Tariff

Table 4-12 provides a comparison of the actual network prices for 2007/08 and the proposed prices for the Subtransmission ToU Demand Tariff for 2008/09.

Component	2007/08		2008/09		% Ch in DUOS	% Ch in NUOS
	DUOS Tariff	NUOS Tariff	DUOS Tariff	NUOS Tariff		
Network Access Charge (\$ pa)	6,453	6,453	6,796	6,796	5.3%	5.3%
Peak Charge (c/kWh)	0.84	1.61	0.89	1.46	6.3%	-9.5
Shoulder Charge (c/kWh)	0.44	1.21	0.44	1.00	1.2%	-17.1
Off-Peak Charge (c/kWh)	0.01	0.28	0.01	0.43	5.8%	53.4
Seasonal Peak Demand (\$/KVA/Month)	4.05	4.05	4.53	4.77	11.9%	17.7
Non Seasonal Peak Demand (\$/KVA/Month)	3.86	3.86	4.31	4.55	11.9%	17.9

Table 4-12: Integral's Proposed 2008/09 ST Time of Use Demand Tariff

4.3 Customers on Locational Network Tariffs

4.3.1 Cost Reflective Network Pricing Customers

Integral's DUOS pricing proposal is to increase individually calculated Cost Reflective Network Tariffs on average by around the overall WAPC increase for 2008/09.

4.3.2 Other Customers

There are also a small number of customers currently on confidential network tariffs (N23 and N24) that do not satisfy the criteria for a cost reflective network tariff. Integral intends to transition these customers to the published High Voltage ToU Demand Network Tariff. This transition will be governed by the price limits set by the Tribunal and Integral's desire to not expose these customers to unacceptable price shocks.

4.4 Miscellaneous Service Charges and Monopoly Service Charges

4.4.1 Miscellaneous Service Charges

Integral's proposed Miscellaneous Service Charges are shown in the following Table.²

Miscellaneous Service	Proposed 2008/09 Charge (Excluding GST)
Special Meter Reading	\$35
Meter Test	\$58
Supply of Conveyancing information – desk inquiry	\$29
Supply of Conveyancing information – field visit	\$58
Off-peak conversion	\$47
Disconnection visit (acceptable payment received)	\$35
Disconnection at meter box	\$70
Disconnection at pole top/pillar box	\$117
Rectification of illegal connection	\$175
Reconnection outside business hours	\$75

Table 4-12: Integral's Proposed 2008/09 Miscellaneous Service Charges

² As per IPART 2004, Final Network Price Determination , Annexure 2, June 2004.

4.4.2 *Monopoly Service Charges*

The proposed charges for miscellaneous services in 2008/09 are consistent with Annexure 2 of the 2004 Network Determination.³ The proposed charges for 2008/09 are provided in Appendix 2.

³ As per IPART 2004, Network Price Determination , Annexure 2, June 2004

5 Compliance with Regulatory Arrangements

Annexure 17 of the 2004 Network Determination requires that the Annual Network Pricing Report submitted by a DNSP under clause 12 must:

Explain how the prices meet the regulatory arrangements, including price limits on network tariffs, weighted average price cap formula, and provide for the recovery of Transmission Related Payments.

5.1 Compliance with Price Limits and overall WAPC Constraint

The following table demonstrates that Integral's 2008/09 network pricing proposal complies with the price limits and the overall X factor constraint outlined in the 2004 Network Determination. A more detailed discussion on compliance with the price limit set by the Tribunal is contained in the Appendix section at the end of this report.

Compliance item	Outcome
WAPC	
Consumer Price Index (All Groups, Weighted Average, 8 Capital Cities)	2.30%
X Factor	1.50%
D Factor	0.10%
WAPC = CPI + X + D	3.90%
Weighted Average Price Change in proposed DUOS Tariff	3.90%
Do the Proposed DUOS tariffs meet the WAPC criterion?	YES
Are total customer numbers and GWh consumption unchanged by reasonable estimates?	YES
PRICE LIMITS	
L Factor	4.50%
CPI + L	6.80%
Fixed Charge Limit	\$30
Are all proposed network tariffs within the price limit?	YES
Are all proposed residential fixed charges within the price limit?	YES
TRANSMISSION RECOVERY	
Forecast closing balance of Transmission overs/(unders) account (\$'000s) as at 30 June 2009	0
Forecast closing balance of Transmission overs/(unders) account as a percentage of TCR revenue as at 30 June 2009	0
Transmission Recovery Amount requiring approval	(0)
APPROVED COST PASS-THROUGH RECOVERY	
General cost pass-through amount	\$23,161
Specific cost pass-through amount	\$1,740
Cost pass-through is not greater than approved amount	YES

Table 5-1: Compliance Worksheet in IPART WAPC Compliance Template

6 Customer Impact of Integral's Proposed Network Tariffs

Annexure 17 of the 2004 Network Determination requires that the Annual Network Pricing Report submitted by a DNSP under clause 12 must:

Demonstrate the impact of the forthcoming change in the network tariffs on typical customer's bills, including disclosing forecast average prices (based on typical bill categories).

This section discusses the customer impacts of Integral's proposed network prices for 2008/09.

6.1 Impact of Proposed Network Prices for 2008/09 – Below 160 MWh pa

6.1.1 Proposed Default Residential Tariff

The following table shows the customer impact of Integral's proposed Default Residential Network tariff for 2008/09.

Annual Consumption	DUOS Bill		NUOS Bill		% Change in DUOS Bill	% Change in NUOS Bill
	2007/08	2008/09	2007/08	2008/09		
0	77	77	77	77	0.0%	0.0%
3,500	265	277	285	306	4.8%	7.4%
6,200	410	432	446	483	5.5%	8.4%
7,500	485	512	529	574	5.6%	8.6%
15,000	969	1026	1057	1150	6.0%	8.9%

Table 6-1: Customer Impact – Integral's Proposed Residential IBT Network Tariff

Integral's 2008/09 pricing proposal for the default residential tariff is expected have the following customer impacts at the NUOS level in 2008/09, as summarised below:

- A small customer consuming 3,500 kWh per annum – the NUOS component of their electricity bill is estimated to increase in nominal terms by no more than approximately 6.6% or \$19 per annum;
- An average customer consuming 6,200 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase in nominal terms by approximately 7.5% or \$33 per annum; and
- A large customer consuming more than 15,000 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase in nominal terms by no more than approximately 8.0% per annum.

6.1.2 Proposed Controlled Load 1 Tariff

The following table shows the customer impact of Integral's proposed prices for Controlled Load 1 Network Tariff for 2008/09.

Annual Consumption	DUOS Bill		NUOS Bill		% Change in DUOS Bill	% Change in NUOS Bill
	2007/08	2008/09	2007/08	2008/09		
0	7.3	7.3	7	7	0%	0%
1,000	9	9	12	12	0%	4%
3,500	14	14	24	25	0%	7%
7,500	22	22	43	46	0%	8%
10,000	26	26	54	59	0%	9%

Table 6.2: Customer Impact – Integral's Proposed Controlled Load 1 Network Tariff

Integral's 2008/09 pricing proposal for the controlled load 1 tariff is expected have the following customer impacts at the NUOS level in 2008/09, as summarised below:

- A small customers consuming less than 1,000 kWh per annum – the NUOS component of their electricity bill is estimated to decrease by -23% in nominal terms in 2008/09;
- An average customer consuming 3,500 kWh per annum – the NUOS component of their annual electricity bill is estimated to decrease in nominal terms by approximately -41% or \$10 per annum in 2008/09; and
- A large customer consuming more than 10,000 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase in nominal terms by no more than approximately -51% in 2008/09.

6.1.3 Proposed Controlled Load 2 Tariff

The following table shows the customer impact of Integral's proposed NUOS prices for Controlled Load 2 Network Tariff for 2008/09.

Annual Consumption	DUOS Bill		NUOS Bill		% Change in DUOS Bill	% Change in NUOS Bill
	2007/08	2008/09	2007/08	2008/09		
0	7	7	7	7	0%	0%
1,000	22	22	26	28	0%	5%
3,500	60	60	73	78	0%	7%
10,000	157	157	195	209	0%	7%

Table 6.3: Customer Impact – Integral's Proposed Controlled Load 2 Network Tariff

Integral's 2008/09 pricing proposal for the controlled load 1 tariff is expected have the following customer impacts at the DUOS level in 2008/09, as summarised below:

- A small customers consuming less than 1,000 kWh per annum – the DUOS component of their electricity bill is estimated to decrease by 4% in nominal terms in 2007/08;
- An average customer consuming 3,500 kWh per annum – the DUOS component of their annual electricity bill is estimated to decrease in nominal terms by approximately 4% or \$3 per annum in 2007/08; and
- A large customer consuming more than 10,000 kWh per annum – the DUOS component of their annual electricity bill is estimated to decrease in nominal terms by no more than approximately 5% in 2007/08.

6.1.4 Proposed Default General Supply Tariff

Annual Consumption	DUOS Bill		NUOS Bill		% Change in DUOS Bill	% Change in NUOS Bill
	2007/08	2008/09	2007/08	2008/09		
0	77	77	77	77	0.0%	0.0%
10,000	509	543	564	611	6.6%	8.3%
20,000	1020	1091	1129	1226	7.0%	8.6%
40,000	2040	2186	2259	2457	7.1%	8.8%
80,000	4081	4375	4519	4919	7.2%	8.8%

Table 6-4: Customer Impact – Proposed Default General Supply Tariff

The impact on general supply customers of Integral's proposed 2008/09 network tariff is summarised below:

- A smaller than average general supply customers consuming less than 10,000 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase by approximately 9.8% or \$55 per annum in nominal terms;
- An average sized customers consuming 20,000 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase in nominal terms by no more than approximately 10.1% or \$114 per annum; and
- A large customers consuming more than 40,000 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase in nominal terms by no more than approximately 10.3% or \$232 per annum; and
- A very large customers consuming more than 80,000 kWh per annum – the NUOS component of their annual electricity bill is estimated to increase in nominal terms by no more than approximately 10.3% per annum.

6.1.5 Proposed Transitional Time of Use Tariff

The following figure shows the customer bill impact distribution (expressed in nominal terms) of Integral's proposed Transitional Time of Use Tariff for 2008/09.

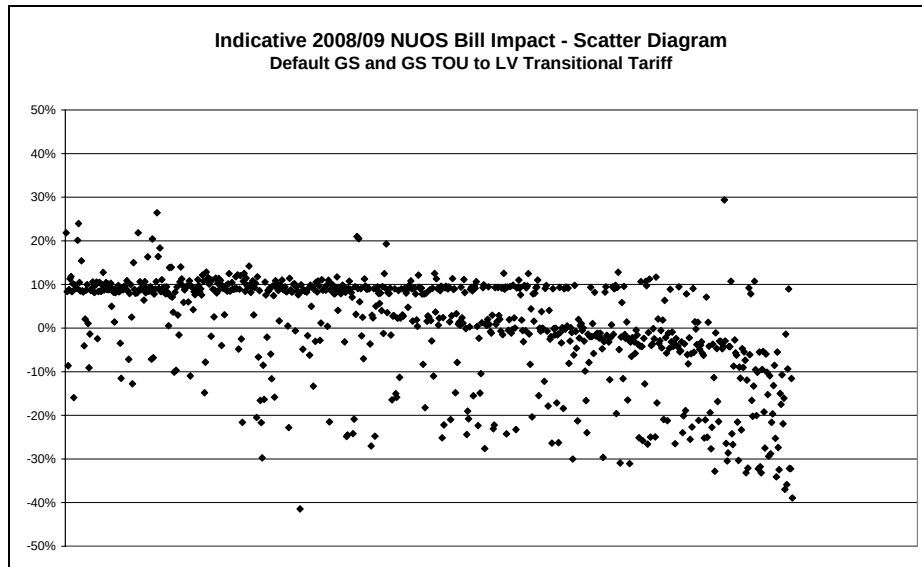


Figure 6-1: Customer Impact (Nominal Terms) – Transitional TOU Network Tariff

6.2 Impact of Proposed Network Prices for 2008/09 – Above 160 Mwh pa

6.2.1 Low Voltage Time of Use Demand Tariff

The following figure shows the customer bill impact distribution (expressed in nominal terms) of Integral's proposed Low Voltage Time of Use Demand Tariff for 2008/09.

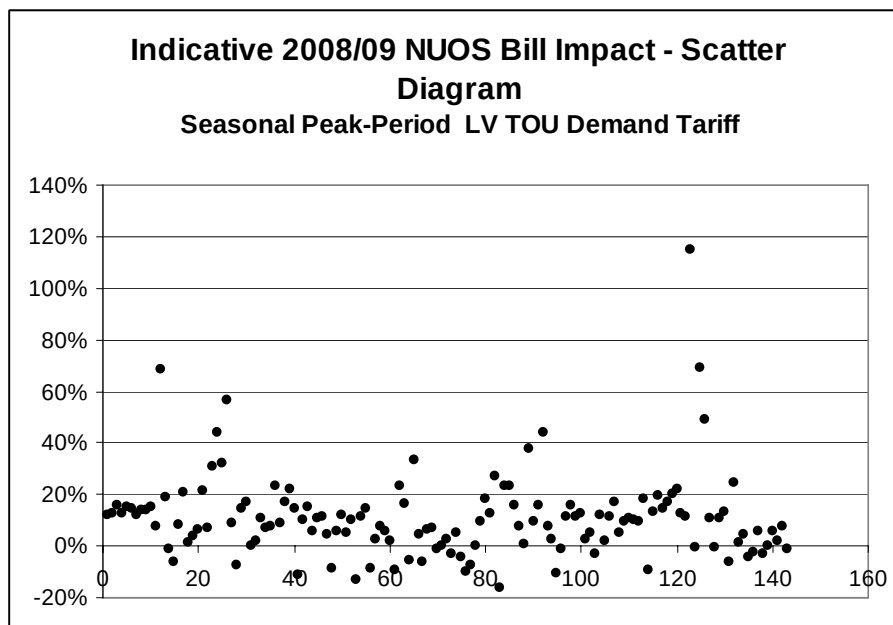


Figure 6-2: Customer Impact (Nominal Terms) – LV TOU Demand Network Tariff

6.2.2 High Voltage Time of Use Demand Tariff

The following figure shows the customer bill impact distribution (expressed in nominal terms) of Integral's proposed High Voltage Time Of Use Demand Tariff for 2008/09.

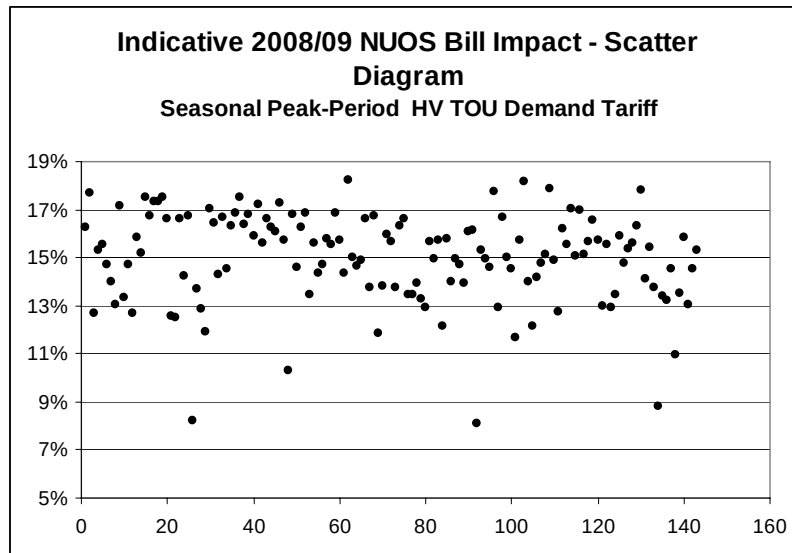


Figure 6-2: Customer Impact (Nominal Terms) – HV TOU Demand Network Tariff

6.2.3 Subtransmission Time of Use Demand Tariff

The following figure shows the customer bill impact distribution (expressed in nominal terms) of Integral's proposed Subtransmission Voltage Time of Use Demand Tariff for 2008/09.

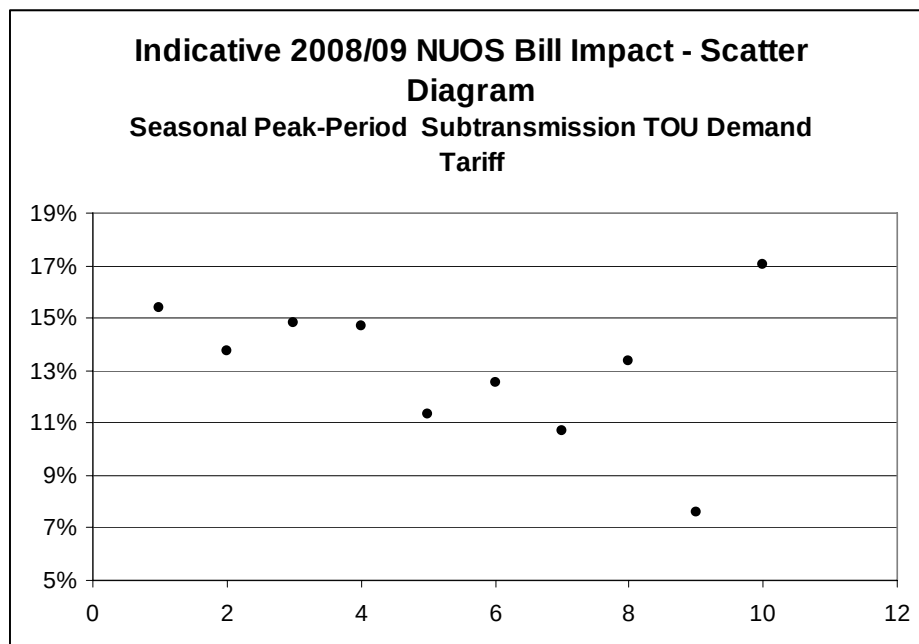


Figure 6-3: Customer Impact (Nominal Terms) – ST TOU Demand Network Tariff

7 Consistency with Pricing Principles and Network Tariff Strategy Statement

Annexure 17 of the 2004 Network Determination requires that the Annual Network Pricing Report submitted by a DNSP under clause 12 must:

Confirm and demonstrate that the prices are consistent with the Pricing Principles and the information contained in the Network Tariff Strategy Statement.

The purpose of this section is to provide stakeholders with an understanding of whether the proposed network prices presented in this report comply with the key IPART's Pricing Principles set out in Annexure 13 of the 2004 Network Determination.

In essence, IPART's pricing principles *"aim to achieve prices for prescribed distribution services that:*

1. *Reflect economic costs by:*
 - *Being subsidy free*
 - *Having regard to the level of available network capacity*
 - *Signalling future network investment costs*
 - *Discouraging uneconomic bypass*
 - *Allowing negotiation to better reflect the economic costs of specific services.*
2. *Return an appropriate revenue stream while recovering the gap between marginal and average costs in the least distorting manner possible.*
3. *Promote equity, stability and consistency of outcomes by:*
 - *Having regard to impact of price changes on customers*
 - *Being transparent*
 - *Being based on published costs and methods."*⁴

7.1 Prices Signal the Economic Cost of Service Provision

The proposed network prices for 2008/09 presented in this report aim to continue to make progress in addressing current pricing inefficiencies and inequities by improving the network price signal from an economic perspective. Integral believes that the proposed changes to Integral's network tariffs in 2008/09, particularly in relation to Integral's TUOS tariffs, will result in a more equitable recovery of distribution network costs from customers and ensure that more customers will make their marginal consumption decisions on the basis of prices that are more reflective of economic cost.

⁴ IPART 2004, NSW Electricity Distribution Pricing 2006/07 to 2008/09, January, p.139

7.1.1 Are Network Tariffs Subsidy Free?

Integral's network tariffs lie within the subsidy free range in that the average price charged to individual customers within each customer class lies between the standalone and incremental costs of supply. Please refer to Integral's Network Tariff Strategy Statement for more information on this issue. This report is available from www.integral.com.au.

7.1.2 Are Network Tariffs Set in Regard to the Level of Available Network Capacity?

Integral's network tariffs reflect the average cost of augmentation of the network and are uniform across Integral's franchise area. When locational congestion and related capital expenditure requirements are significant, customers receive price signals through connection charges and requirements to contribute connection assets in accordance with Integral's Customer Connection Policy.

7.1.3 Do Network Tariffs Signal Future Network Investment Costs?

Integral's network tariffs aim to signal future network investment costs within the limitation of existing metrology and to facilitate a wider adoption of more efficient network tariffs by Integral's customers. The significant tariff reforms undertaken over recent years have enhanced the economic efficiency of Integral's network price signals.

Under the IBT structure, Integral intends to transition the second block price of the default residential tariff and the default general supply tariff to the level of Long Run Incremental Cost (LRIC) over the longer term.

Integral commissioned Charles River Associates to assess the economic and financial implications of air conditioning use on Integral's distribution network using a LRIC framework. One of major findings of this study is that LRIC for Integral lies between a lower bound of between approximately 8 cents per kWh of energy consumed to an upper bound of approximately 20 cents per kWh of energy consumed. While Integral believes that further work is required in this area, it is, nevertheless, clear that Integral's current variable energy charge for small customers does not properly signal the impact of additional usage on future investment costs. The LRIC calculation methodology used by CRA is explained in Appendix 4.

The IBT structure will result in the variable component (avoidable) being more reflective of economic costs. This will ensure that large users that contribute to the need to invest in peak network capacity being exposed to a higher marginal price. Importantly, large users that do not contribute to the need to invest in peak network capacity will have the incentive to switch from the default tariff to the residential ToU Tariff.

In addition, Integral intends to improve the price signal for interval metered customers by transitioning peak energy and seasonal peak maximum demand charges to a level that is more closely reflective of forward looking economic costs.

7.1.4 Do Network Tariffs Discourage Uneconomic Bypass and Allow Negotiation to Better Reflect the Economic Value of Specific Services?

In practice, only the largest customers have the incentive to bypass the distribution system. Integral undertakes individual distribution cost of supply analysis for all large

customers on individually calculated network tariffs to ensure that prices are cost reflective and do not encourage uneconomic bypass.

Integral's current threshold of a maximum monthly demand of 10MVA or consumption of 40 GWh per annum has resulted in more customers receiving individually calculated network tariffs, which will further discourage uneconomic bypass from occurring.

In situations where negotiation could occur, such as where a generator is requiring connection to the distribution network, Integral will negotiate specific charges with customers where necessary to ensure an economically efficient outcome.

7.1.5 Minimise Distortion of Customers' Consumption and Investment Decisions

Integral is committed to improving the network price signal from an economic perspective. In practice, this commitment means that Integral aims to set the levels of the variable components of network tariffs consistent with economic costs within the limitations of the associated metrology. The level of fixed component of network tariffs is set to enable Integral to recover the gap between marginal and average costs for that particular tariff class. This approach means that Integral is able to recover an appropriate revenue stream in a manner that minimises the loss of economic efficiency (consumption and network investment) from pricing imperfections.

7.2 Promote Equity, Stability and Consistency

Integral understands that the pursuit of network tariff reform has the potential to result in some customers experiencing significant adjustment costs. To ensure that an appropriate balance is struck between economic efficiency, equity and revenue sufficiency, Integral has:

- Taken a gradual approach to reforming the IBT price signal, particularly in relation to increasing the price difference between the two consumption blocks;
- Introduced a new residential ToU Tariff based on a less expensive Type 6 meter. It is expected that this voluntary tariff option will lead to a more cost reflective outcome for the large domestic customers with a significant proportion of their energy use in off-peak periods.
- Safeguarded the interests of customers by implementing transitional arrangements in relation to the following:
 - Integral's policy of compulsory assignment of customers above 160 MWh per annum on to the LV Demand ToU Tariff.
 - Integral's decision to move customers currently on confidential network tariffs to an appropriate standard network tariff, where it is economically desirable to do so.
- Integral has strived to improve the robustness of its Distribution Cost of Supply Model to ensure that distribution network costs are allocated properly to individual tariff classes.

7.3 Consistency with Network Tariff Strategy Statement

The proposed changes to the level and structure of Integral's network tariffs, as set out in this report is consistent with the information contained in Integral's Network Tariff Strategy Statement.

Integral's Network Tariff Strategy Statement is available to be downloaded from www.integral.com.au

8 Expected Levels of Service

Annexure 17 of the 2004 Network Determination requires that the Annual Network Pricing Report submitted by a DNSP under clause 12 must:

Outline the expected levels of service.

This section summarises the expected service levels for the 2004 Determination period. However, it is important to note that the expected levels of service are expressed in two dimensions:

- Network reliability; and
- Our interaction with our customers.

8.1 Network Reliability

The reliability of the electricity supply is one of the key customer service measures for electricity network operators. Network reliability performance is impacted by a variety of factors, many of these factors are largely external to the network itself and therefore beyond the control of the network operator. The cause of supply failure can include adverse weather conditions, defective equipment, foreign interference, human element, lightning, loss of transmission supply and tree contact.

8.1.1 Recent Trend in Reliability Performance

The performance of Integral's network in terms of the System Average Interruption Duration Index (SAIDI) is shown in the following figure.

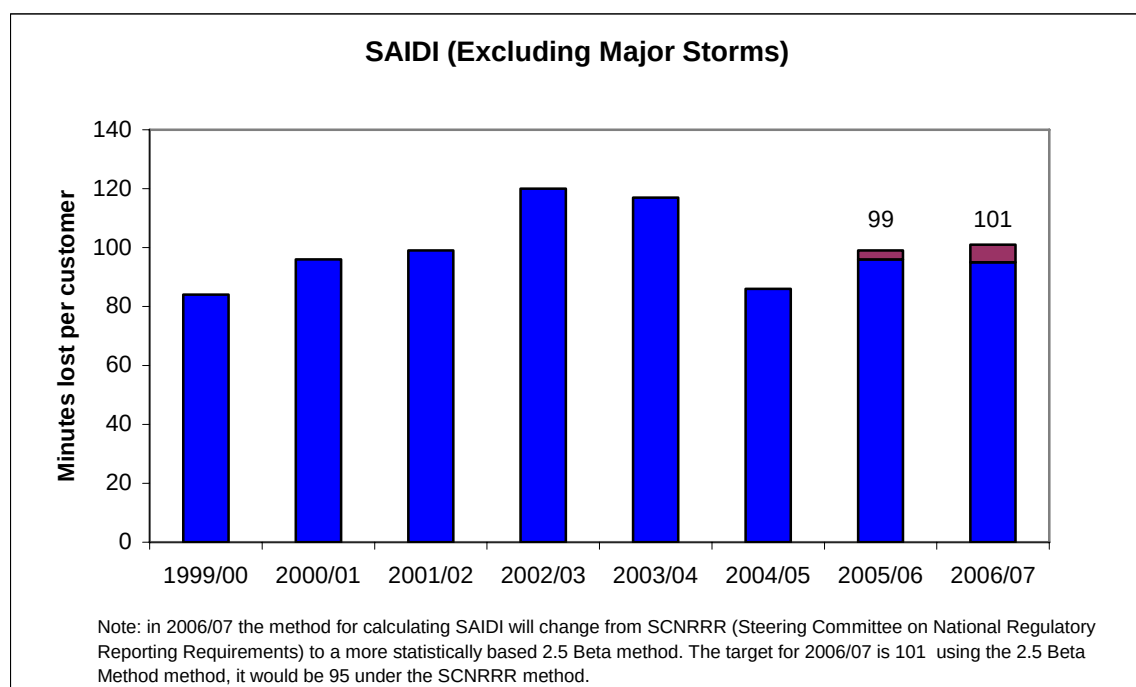


Figure 8.1: Integral Energy Reliability Performance - SAIDI

As shown the above figure, a dramatic improvement in SAIDI occurred in 2004/05, due, in part, to the impact of continuing settled weather conditions. The improving trend in the underlying reliability of the network has been achieved by focussing on improving the reliability of poor performing feeders, continued management of the load at risk and addressing the weighted average life of Integral's network assets.

8.1.2 Reliability by Locality

Not all parts of Integral's network experience similar reliability.

The topography of the network is a fundamental driver of network reliability performance. Given that Integral's network is characterised predominantly as an overhead network, external influences such as adverse weather, bushfire, vegetation impact and vehicle impact are likely to have a much greater influence on network reliability compared to a network that is predominantly underground. Consequently, there is a certain amount of volatility in individual feeder reliability performance attributed to local environmental conditions.

Integral Energy publishes its annual reliability results by local government area in our Network Performance Report as submitted to the Department of Energy, Utilities and Sustainability. This report can be found on Integral Energy's web site (www.integral.com.au).

8.1.3 Network Reliability Strategy

Integral's Network Reliability Strategy establishes a prudent level of capital expenditure and a range of operational response activities to address reliability performance. The reliability strategies developed include:

- Improving the performance of the poor performing lines throughout the network where these deviate significantly from typical performance;
- Reducing the number and duration of interruptions to supply for planned works;
- Addressing the issues which are causing significant contribution to lost customer minutes by the subtransmission network;
- Reducing the susceptibility of the network to damage during major storm events; and
- Improving response times to failures so that the duration of outages is reduced.

8.2 License Conditions

On 1 August 2005, the Minister for Energy and Utilities imposed reliability performance Licence Conditions on all New South Wales Distribution Network Service Providers. These Licence Conditions cover three main areas:

- A standard is set for Integral's average performance for unplanned SAIDI and SAIFI in 2005/06 with annual reductions for the next 6 years. This average is separated into feeder types of Rural Short and Urban with minimum standard performance levels for each.

- Individual feeder standards to set a threshold for identifying poor performing feeders, with a requirement for investigating, plan development and reporting on progress of these feeders.
- Customer service standards are also set for interruption duration and frequency in metropolitan and non-metropolitan areas.

For Integral, the limits are shown in the following table:

SAIDI (Minutes Per Customer) – Integral Average						
	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11
Urban	90	88	86	84	82	80
Short Rural	300	292	284	276	268	260
SAIFI Stand (Number/Customer) – Integral Average						
	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11
Urban	1.3	1.28	1.26	1.24	1.22	1.2
Short Rural	2.8	2.76	2.72	2.68	2.64	2.6

Table 8-1: Integral's Licence Conditions Reliability Standards – Average Performance

For the performance of individual feeders, the limits are shown in the following table:

	SAIDI (Minutes Per Customer)	SAIFI Stand (Number/Customer)
Urban	350	4
Short Rural	800	6.5

Table 8-2: Integral's Licence Conditions Reliability Standards – Individual Feeder Performance

- *Our Interaction with Our Customers*

The following service standards form part of the Standard Form Customer Connection Contract for Integral:

- Timely provision of services
- Timely notice of planned interruptions to electricity supply
- Provision of telephone hotlines
- Punctuality in keeping appointments
- No disconnection on weekends or public holidays
- No disconnection except after due notice
- Notice to be provided after disconnection

- Disconnection at your request
- Time limit for new connection services
- Repair of faulty street lighting

More detailed information on Integral's Guaranteed Customer Service Standards for specific aspects of our interaction with our customers is contained in Integral's Standard Form Customer Connection Contract, which is available from www.integral.com.au.

9 Project Capital Expenditure

Annexure 17 of the 2004 Network Determination requires that the Annual Network Pricing Report submitted by a DNSP under clause 12 must:

Outline the project capital expenditure projects that will occur for the coming year

This section summarises Integral's proposed capital expenditure over the current regulatory period.

Program	2004/05	2005/06	2006/07	2007/08	2008/09
Replacement – end of life	82.2	74.9	63.3	73.3	83.7
Growth(demand related)	108.8	140.1	133.4	126.4	92.5
Reliability and quality improvement	18.4	20.6	19.9	20.3	18.5
Environment, safety, stat., other	19.1	17.5	15.7	14.2	2.2
Metering	5.0	5.8	6.2	6.5	2.2
Total	233.4	259.0	238.4	240.7	199.0

Table 9-1: Integral's Projected Annual Capital Expenditure – 2004 Network Determination

Please note that more detailed information about Integral's proposed capital expenditure is available in the Network 2017 publication, which is available from www.integral.com.au.

It is also important to note that IPART approved additional capital expenditure for Integral Energy under the cost pass through provisions of the 2004 Network Determination in respect to the new distribution license conditions relating to reliability performance, as shown below:

CAPEX (\$ 2005/06 million)	2004/05	2005/06	2006/07	2007/08	2008/09
General cost pass through	-41.6 ⁵	3.5	91.3	94.8	199.8
Specific cost pass through	0.0	0.0	0.5	0.0	0.0
Total	-41.6	3.5	91.8	94.8	199.8

Table 9-2: Integral's Approved Capital Expenditure – Cost Pass Through

⁵ Note - Negative capital expenditure in 2004/05 is to back out capital expenditure that formed part of the 2004 Determination and contributed to meeting new license conditions.

10 Forecast Electricity Consumption and Maximum Demand

Annexure 17 of the 2004 Network Determination requires that the Annual Pricing Report submitted by a DNSP under clause 12 must:

Outline the expected consumption for major customer classes and maximum demand for the coming year.

This section summarises Integral's current consumption and maximum demand forecast in 2008/09 for each major customer class.

10.1 Integral's 2007/08 forecasts for electricity consumption (MWh) and maximum demand (MVA) per tariff class

Network Tariff	Total (MWh)	Non - TOU (MWh)	TOU (MWh)	Seasonal Peak Demand (kVA)	Non-Seasonal Peak Demand (kVA)
Residential Tariffs	4,840,261	4,827,014	13,247		
Controlled Load Tariffs	955,578	955,578			
General Supply Tariffs	2,292,353	1,751,160	541,193		
LV ToU Demand Tariff	3,357,635		3,357,635	6,321,790	3,022,940
HV ToU Demand Tariff	1,174,199		1,174,199	2,157,372	1,048,218
ST ToU Demand Tariff	458,530		458,530	834,148	429,971
Other (CRNP, IDTs & Streetlighting)	4,087,470	99,444	3,988,026	5,958,633	3,111,859
Total	17,166,026	7,633,196	9,532,830	15,271,943	7,612,988

Table 10-1: Integral's volume forecasts for 2008/09, by tariff class

10.2 Integral's 2008/09 forecasts of number of customers per tariff class

Network Tariff	Number of customers
Residential Tariffs	770,798
Controlled Load Tariffs	321,259
General Supply Tariffs	76,447
LV ToU Demand Tariff	3,354
HV ToU Demand Tariff	244
ST ToU Demand Tariff	29
Other (CRNP Tariffs, IDTs & Streetlighting)	42
Total	1,172,173

Table 10-2: Integral's customer number forecasts for 2008/09, by tariff class

Appendix 1: Monopoly Service Charges with Contestable Work

The proposed charges for miscellaneous services in 2007/08, which are consistent with Clause 4 of the 2004 Network Determination, are outlined below:

Monopoly Service	Underground urban residential subdivision				Rural Overhead Subdivisions and Rural Extensions				Underground Commercial and Industrial or Rural Subdivisions				Commercial and Industrial Development	Asset Relocation or Street Lighting
Design Information	Up to 5 lots \$126 6 to 10 lots \$189 11 - 40 lots \$315 Over 40 lots \$378				R2 per hour				R2 per hour				R2 per hour	R2 or R3 per hour
Design Certification	Up to 5 lots \$63 6 to 10 lots \$126 11 - 40 lots \$189 Over 40 lots \$252				1 - 5 poles \$63 6 -10 poles \$126 11 or more poles \$189				Up to 10 lots \$126 11 - 40 lots \$189 Over 40 lots \$378				R3 per hour	R2 or R3 per hour
Design Rechecking	R2 per hour				R2 per hour				R2 per hour				R3 per hour	R2 or R3 per hour
Inspection Fee	Grade:	A	B	C	Grade:	A	B	C	Grade:	A	B	C	R2 or R3 per hour	R2 or R3 per hour
	per lot	per lot	per lot		per pole	per pole	per pole		per lot	per lot	per lot			
	First 10 lots:	\$32	\$76	\$158	1-5 poles:	\$38	\$76	\$139	First 10 lots:	\$32	\$76	\$158		
	Next 40 lots:	\$19	\$44	\$95	6-10 poles:	\$32	\$63	\$126	Next 40 lots:	\$32	\$76	\$158		
	Remainder:	\$6	\$25	\$44	11+ poles:	\$25	\$44	\$95	Remainder:	\$32	\$76	\$158		

Access Permit	Residential Subdivisions: \$21.00 per lot combined fee	\$935 max. per access permit	\$935 max. per access permit	\$935 max. per access permit	\$935 max. per access permit
Substation Commissioning		\$701 per substation	\$701 per substation	\$701 per substation	\$701 per substation
Monopoly Service	Underground urban residential subdivision	Rural Overhead Subdivisions and Rural Extensions	Underground Commercial and Industrial or Rural Subdivisions	Commercial and Industrial Development	Asset Relocation or Street Lighting
Adminis-tration	Up to 5 lots \$153 6 - 10 lots \$204 11 - 40 lots \$255 Over 40 lots \$306	Up to 5 poles: \$153 6-10 poles: \$204 11 or more poles \$306	R1 per hour (max 6 hours)	R1 per hour (max 6 hours)	R1 per hour
Notice of Arrangement	\$153				
Re-Inspection	R2 per hour (max 1 hour per level 2 reinspection)				
Access	R1 per hour				
Authori-sation	\$126				
Inspection of Service Work (Level 2 work)	All Service connections: A Grade : \$16 per NOSW B Grade: \$26 per NOSW C Grade: \$76 per NOSW (NOSW = Notification of Service Work)				
Site Establishment	\$110				

Table A1-1: Summary of Integral's proposed 2007/08 Monopoly Service Charges

Appendix 2: Compliance with Price Limits

Price Limit on movements in Individual Network Tariffs

The IPART Network Price Determination has set limits on movements in individual network tariffs such that the weighted average increase in the tariff components is not able to exceed the specified price limit. As outlined in Annexure 9 of the IPART Network Price Determination the Tribunal has set the NUOS price limit for Integral to CPI+4.5% for 2008/09

The following Table demonstrates that Integral's proposed NUOS tariffs in 2006/07 satisfy the Tribunal's price limit.

Network Tariff	Proposed 2008/09 % Ch (nominal)	Price Limit	Compliant (Yes/No)
Default Residential Tariff	6.8%	6.8%	Yes
Residential ToU (Type 5) Tariff	6.8%	6.8%	Yes
Residential ToU Tariff	6.8%	6.8%	Yes
Controlled Load 1 Tariff	6.8%	6.8%	Yes
Controlled Load 2 Tariff	6.8%	6.8%	Yes
Default General Supply Tariff	6.8%	6.8%	Yes
General Supply ToU Tariff	6.8%	6.8%	Yes
LV ToU Demand Tariff	6.8%	6.8%	Yes
HV ToU Demand Tariff	6.8%	6.8%	Yes
ST ToU Demand Tariff	6.8%	6.8%	Yes

Table A2-1: Compliance with NUOS Price Limit at an Individual Tariff Level

Limit on the movement in the fixed component of any residential network tariffs

The IPART Network Price Determination also places an upper limit on increases in the aggregate fixed charge component of any network tariff for residential customers. The following table demonstrates the Integral pricing proposals for residential customers satisfies this limit.

Network Tariff	2006/07 fixed charge	Proposed 2007/08 fixed charge	Change (\$)	Compliant (Yes/No)
Default Residential Tariff	76.65	76.65	0	Yes
Residential ToU(Type 5) Tariff	200.75	208.05	7.30	Yes
Residential ToU Tariff	83.95	113.95	30	Yes

Table A2-2: Compliance with Limit on Residential Fixed Charge Increase

Appendix 3: Long Run Economic Cost Analysis

Integral Energy commissioned Charles River Associates (CRA) to examine the economic and financial implications of air conditioning use on Integral's distribution network. Charles River Associates used a long run incremental (or marginal) cost approach at a network level to provide:

- A means of demonstrating the magnitude of the effect of air conditioning on peak demand;
- The real cost of marginal capacity using an energy consumption basis; and
- The level of cross subsidy of air conditioning gives rise to under the current tariff arrangements.

Long Run Incremental Cost

(i) Charles Rivers Associates (CRA) Study

At the network level, the Long-Run Incremental Cost concept is relatively straight forward. Long-Run Incremental Cost is defined as the total annualised cost (including all allowable returns to the network) of providing and servicing a defined additional increment of network capacity. Embedded in this concept is the proviso that the network operator be able to recover the relevant costs from those customers whose consumption has driven the need for incremental investment in capacity.

The key inputs to methodology adopted by CRA to estimated Long-Run Incremental Cost are summarised below:

- Defined drivers of investment in peak capacity for given locations;
- A calculation of the magnitude of the cost imposed on the network; and
- Evaluation of the difference between that cost and the revenue associated with electricity consumption during the peak period under current network tariffs.

The methodology provides revealing insights, as described below:

- Based on current tariffs and load duration curve for the network (or for sub-parts of the network) it is possible to evaluate how much of the peak load is uneconomic. If the increment of cost associated with the load is not covered by the increment of revenue then it is "uneconomic" and is causing either a cross subsidy or a relative loss or both;
- Pricing to recover the gross amount of revenue required from peak period load would significantly reduce the average charges for the load remaining, thereby reducing any cross subsidy that may exist;
- Forward projections of capital and revenue requirements can be made on the basis of capacity investments that are driven by peak demand, as can some validation or previous expenditures; and

- Costing of demand management initiatives becomes relatively easy as the network becomes indifferent to revenue recovery or the removal of the incremental costs at peak.

The modelling undertaken by CRA for Integral Energy:

- Examined the capital cost of an incremental increase of 1 kW of capacity placed on the system (A kW basis was used instead of kVA basis for simplicity);
- Examined the relative level of this incremental capital cost on the basis that it was sensitive to demand growth in peak times – a range was used for this purpose. This modelling yields a cost per kW per additional increment of peak capacity investment in the network;
- Examined the long run marginal (annual) cost for an additional kW increment investment in the network (average across the network) based on the requirement that a return on the incremental addition to the asset was received by the network operator. Operating Costs, Depreciation and WACC percentages were applied to Integral's 1998 ORC value for this calculation;
- Recognised that for full cost recovery from the peak demand (and hence no potential for a cross subsidy) it is necessary for the network to achieve revenues that equate to this long run marginal cost. If there is a revenue shortfall the amount has to be made up through an effective loading of these costs on the averaged tariff paid by all customers. This is a sub optimal economic outcome that punishes both those consumers who do not drive peak costs and the network operator;
- Used the marginal price of the residential tariff to calculate how long a kW of capacity would have to be used in any given year to actually yield the required revenue;
- Used the load duration curve for the system to determine the point on the curve where revenue shortfall occurs. CRA could then calculate the area above this point to determine the magnitude of the under recovery (and potential subsidised consumption) in total; and
- Calculated the under recovery (and potential subsidy) as the difference in unearned marginal revenue from peak demand and the actual marginal revenue earned.

CRA estimates the range of under recovery/potential cross subsidy at \$80 to \$110 million per annum. The proportion of network replacement costs that are peak load dependent are assumed to fall within the range of 40 to 60%.

ICB at 1998 (DORC)	\$	1,756	million
ORC – 1998 GHD	\$	3,827	million
Network Peak Demand 2000/01		2966	MW
Average Cost/kW to Replace Network	\$	1,290	
Proportion which is Peak Load Dependent		48%	
Marginal Cost for Peak Demand	\$	619	
WACC		7.50%	of Network Value
Depreciation (Peak Load Dependent Assets) 1998		1.75%	of Network Value
Operating Costs (Network Only) 2001		4.31%	of Network Value
		13.56%	Total
Annuity driven marginal cost to replace network ⁶⁷		\$91.15	per kW/year
Marginal revenues from key groups driving demand	\$	0.0420	kWh
Average hours for 1kW of new peak has to be used to meet revenue target using marginal price		2,170	h/year
Load Duration Curve – load at 2,170 hours		510,702	MWh
Marginal Revenue	\$	21	million
MW of Peak load		1,208	MW
Costs to build this	\$	748.16	million
Should get revenues of	\$	110.10	million/year
Under Recovery / Potential Subsidy	\$	88.66	million

Table A3.1 – Example of LRIC Calculation Methodology - Charles River Associates

⁶ The annuity is over a 20-year time frame – extensions to this make little difference. The marginal revenues are based on (the \$0.0420/kWh energy rate) current marginal energy rate for residential customers - the key groups driving peak demand.

⁷ This is the revenue needed to cover costs for peak kW of load

Charles River Associates modelling suggests that:

- If the under recovered costs were smeared over the peak energy consumption responsible for driving those costs, the average price for peak energy consumption would be approximately 20 cents/kWh – some 4 to 5 times the averaged marginal energy rates currently charged; and
- Similarly, if the under recovered costs were smeared over total energy consumption (not solely peak consumption), the average price would need to be increased by approximately 1.5 to 2.0 cents/kWh – 30% above the averaged marginal energy.

(ii) National Economic Research Associates (NERA) Study

In addition to the research undertaken by CRA, Integral Energy also commissioned National Economic Research Associates (NERA) to develop a Long Run Incremental (or Marginal) Cost (LRMC) model for Integral's network and to advise Integral on how to apply these estimates of economic cost to improve the efficiency of Integral's network tariff.

NERA estimated Integral's long-run marginal costs per kVA by using load data sourced from Integral planning documents, expenditure planning data from the Strategic Asset Management Program (SAMP) and accounting and financial data supplied by Integral. The following schematic illustrates how NERA has used these data to arrive at an estimate of Integral's long run marginal cost. The process has been divided into five separate steps.

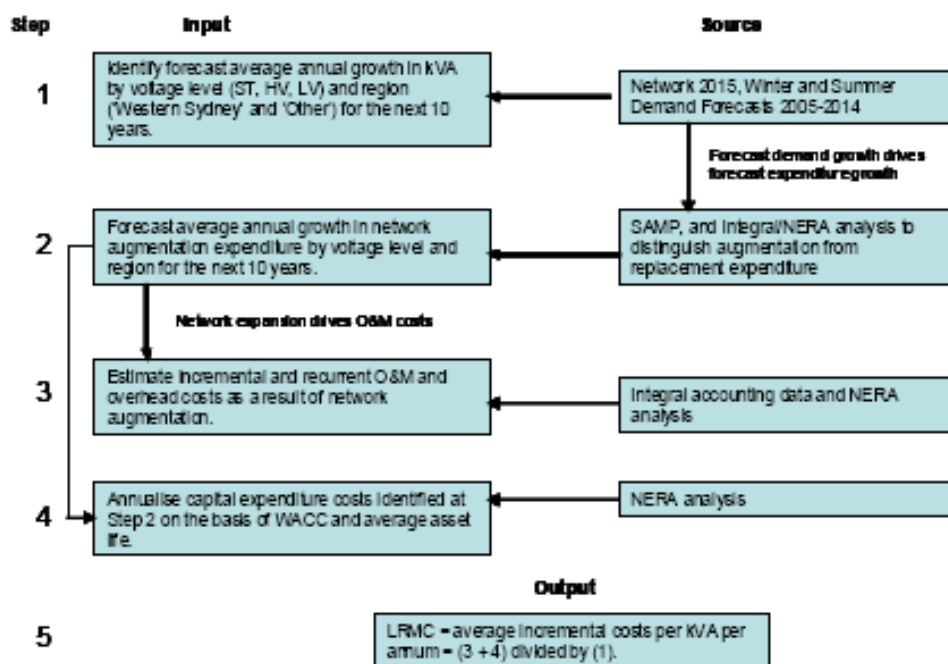


Figure A.3: LRIC Methodology - National Economic Research Associates

The following LRMC estimates represent the cost associated with meeting forecast growth in critical peak demand for apparent energy (kVA). Critical peak demand is any level of demand that, should it occur often enough, would trigger an increase in network augmentation under Integral's network planning strategy.

Voltage Level	LRMC Estimate (\$/kVA)
Low Voltage	113.78
High Voltage	32.39
Subtransmission Voltage	23.57

Table A3.2 - Integral's LRIC – National Economic Research Associates

It should be noted that the values for low voltage and high voltage already include the long run marginal cost of upstream (higher voltage) costs caused by consumption at the lower voltage level.